



AI Infrastructure

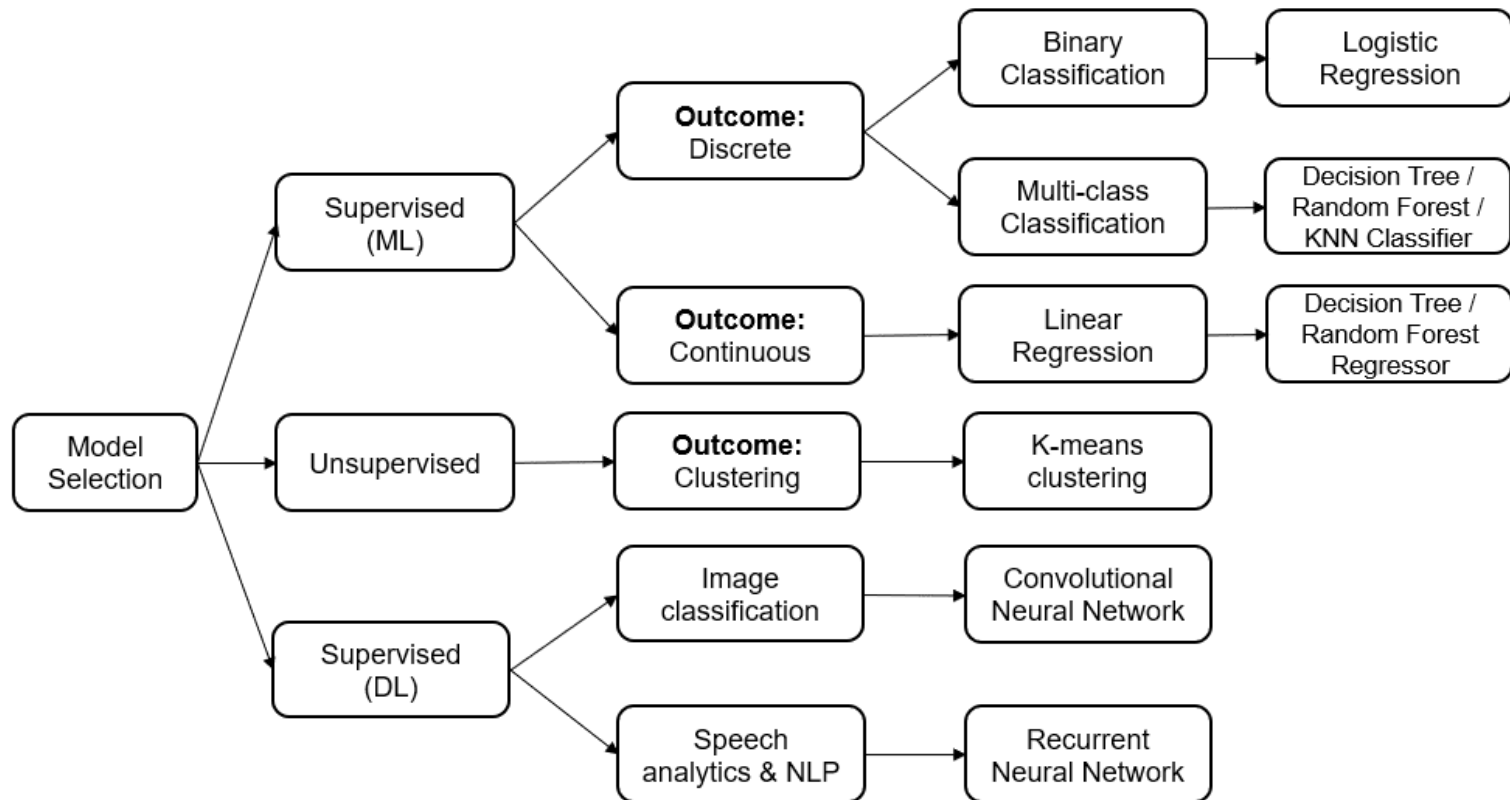
Ioannis Papapanagiotou

ML Model Building

Building of ML model

- **Building an ML Model** requires splitting of data into two sets, such as 'training set' and 'testing set' in the ratio of 80:20 or 70:30;
- A set of supervised (for labelled data) and unsupervised (for unlabeled data) algorithms are available to choose from depending on the nature of input data and business outcome to predict.
- The model is built using training dataset and make prediction using test dataset.

Choosing Model

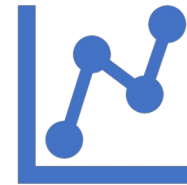


Model Performance

- **Computation of Model Performance** is next logical step to choose the right model.
- The model performance metrics will decide on final model selection, that include computation of accuracy, precision, recall, F1 score (weighted average of precision and recall) , along with confusion matrix for classification models and coefficient of determination for regression models.
- It is not recommended to use accuracy as a measure to determine the performance of classification models that are trained with imbalanced/skewed datasets, rather precision and recall are recommended to be computed to choose right classification model.

Model Development Best Practices

1. Perform clear hypothesis for the identified business problem before attributes identification itself;
2. Build the model using a basic algorithm first, like a logistic regression or decision tree and compile performance metrics, that gives enough confidence about relevance of data before going for a fancier algorithms, like neural networks;
3. Keep intermediate checkpoints in building the model to keep track of its model hyperparameters and its associated performance metrics that gives an ability to train the model incrementally and make good judgement when it comes to performance vs training time;
4. Use real-world production data for training the model to improve correctness of predictions.



What is hyperparameter tuning?

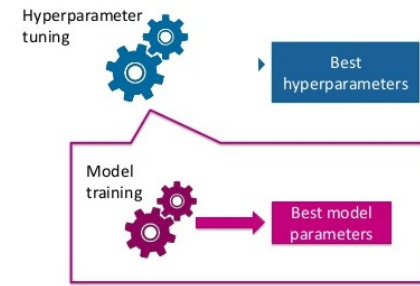
- When you're training machine learning models, each dataset and model needs a different set of hyperparameters, which are a kind of variable.
- The only way to determine these is through multiple experiments, where you pick a set of hyperparameters and run them through your model.
- This is called *hyperparameter tuning*. In essence, you're training your model sequentially with different sets of hyperparameters. This process can be manual, or you can pick one of several automated hyperparameter tuning methods.
- Whichever method you use, you need to track the results of your experiments. You'll have to apply some form of statistical analysis, such as the loss function, to determine which set of hyperparameters gives the best result. Hyperparameter tuning is an important and computationally intensive process.



What are the Hyperparameters

- Hyperparameters are external configuration variables that data scientists use to manage machine learning model training. Sometimes called *model hyperparameters*, the hyperparameters are manually set before training a model.
- They're different from parameters, which are internal parameters automatically derived during the learning process and not set by data scientists.
- Examples of hyperparameters include the number of nodes and layers in a neural network and the number of branches in a decision tree.
- Hyperparameters determine key features such as model architecture, learning rate, and model complexity.

Hyperparameter tuning vs. model training



Why is hyperparameter tuning important?

- Hyperparameters directly control model structure, function, and performance. Hyperparameter tuning allows data scientists to tweak model performance for optimal results. This process is an essential part of [machine learning](#), and choosing appropriate hyperparameter values is crucial for success.
- For example, assume you're using the learning rate of the model as a hyperparameter. If the value is too high, the model may converge too quickly with suboptimal results. Whereas if the rate is too low, training takes too long and results may not converge. A good and balanced choice of hyperparameters results in accurate models and excellent model performance.



Examples of Hyperparameters

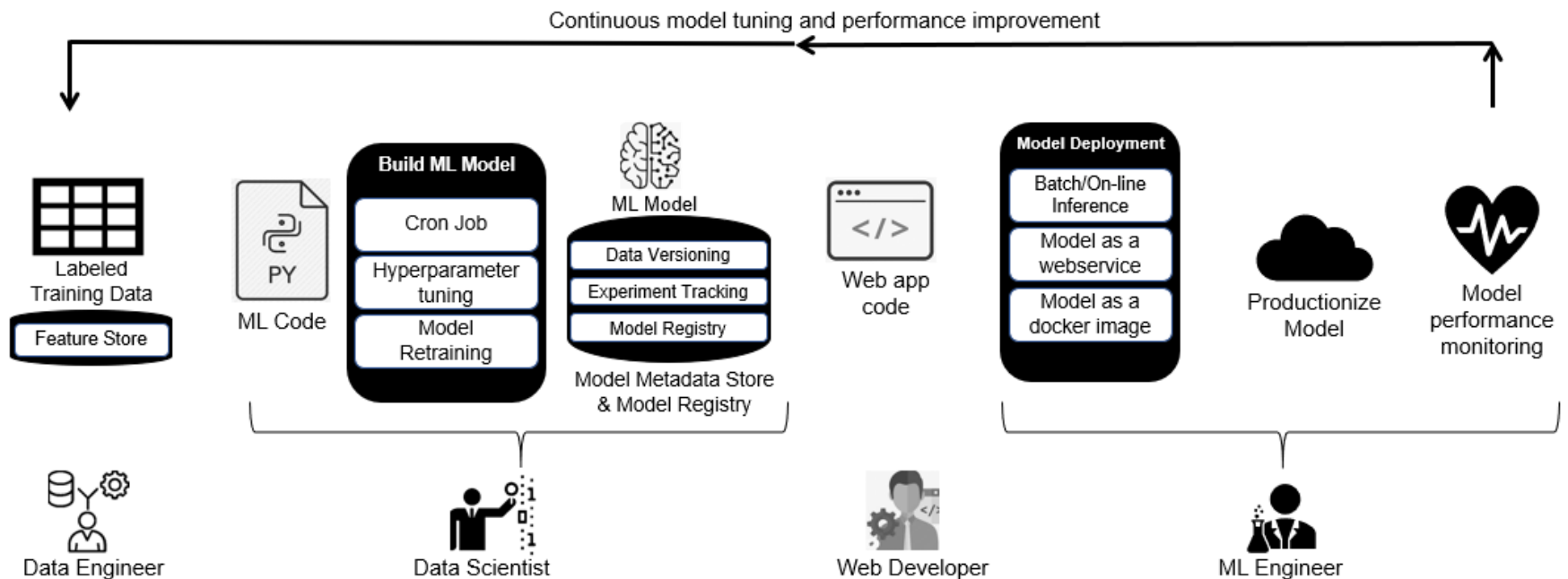
- While some hyperparameters are common, in practice you'll find that algorithms use specific sets of hyperparameters. For example, you can [read how Amazon SageMaker can use image classification hyperparameters](#) and [read how SageMaker can use XGBoost algorithm hyperparameters](#).
- Here are some examples of common hyperparameters:
 - *Learning rate* is the rate at which an algorithm updates estimates
 - *Learning rate decay* is a gradual reduction in the learning rate over time to speed up learning
 - *Momentum* is the direction of the next step with respect to the previous step
 - *Neural network nodes* refers to the number of nodes in each hidden layer
 - *Neural network layers* refers to the number of hidden layers in a neural network
 - *Mini-batch size* is training data batch size
 - *Epochs* is the number of times the entire training dataset is shown to the network during training
 - *Eta* is step size shrinkage to prevent overfitting

MLOps

- **Machine Learning (ML) Model Operations** refers to implementation of processes to maintain the ML models in production environments.
- The common challenge encountered in a typical enterprise scenario is that the ML models worked in lab environment will remain stay at the proof-of-concept stage in many cases.
 - If the model is rolled out into production, it becomes stale due to frequent source data changes that requires rebuilding of model.
- As the models are retrained multiple times, it is required to keep track for model performance and corresponding features and hyperparameters that are used for retraining the model.
- To perform all these operations, there should be a well-defined reproducible process in-place to implement the end-to-end machine learning operations (MLOps) that keeps the model current and accurate in production environment.

ML model Operations Lifecycle

- We cover the entire process of model development to model deployment to model performance monitoring in a seamless manner.



ML Operations

- The model retraining is very important at regular intervals, say fortnightly or monthly or quarterly or on-demand basis as it is very likely that the underlying source data will change over a period in the real-world scenario.
- A cron job is scheduled to retrain the model at the predefined intervals, or as and when the source data is changed, or as and when the model performance is degraded.
- There could be some changes to the model code due to hyperparameter tuning during model re-training.
- It is required to automate these model operations by having an automated model pipeline.